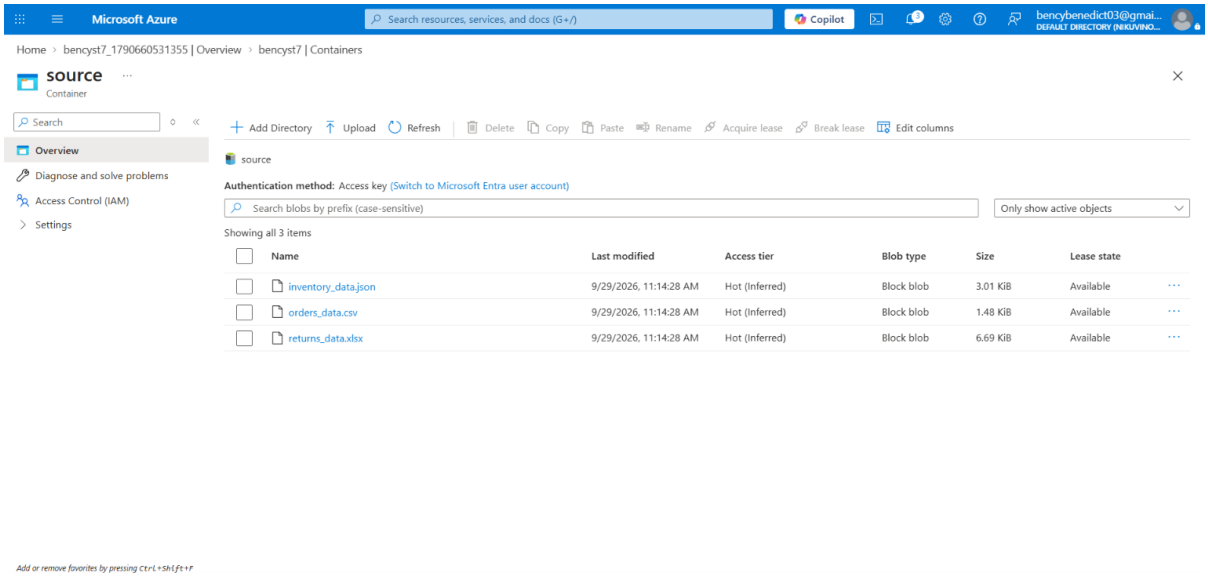
MICROSOFT FABRIC - RETAIL PROJECT

Project Documentation

Goal: copy retail data (CSV, JSON, Excel) from Azure storage into the Bronze layer of a Fabric Lakehouse, clean it in the Silver layer, and build KPIs in the Gold layer.

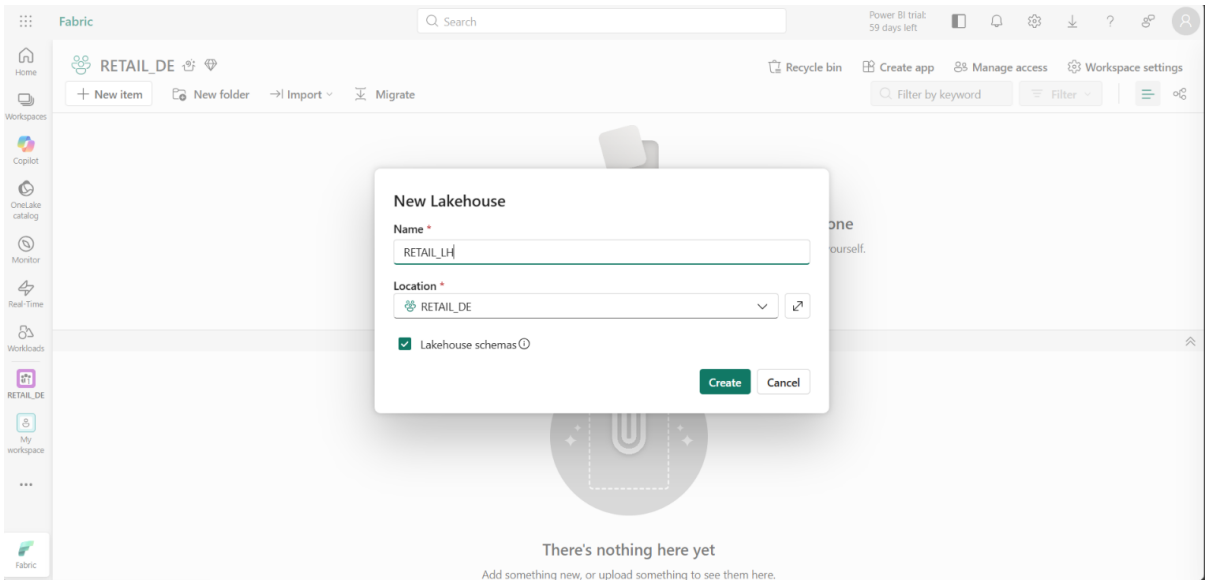
Step 1: Source Files in Azure Storage

- The three source files are stored in the **source** container of an Azure Data Lake Storage Gen2 account.
- Files: **orders_data.csv**, **inventory_data.json** and **returns_data.xlsx**.



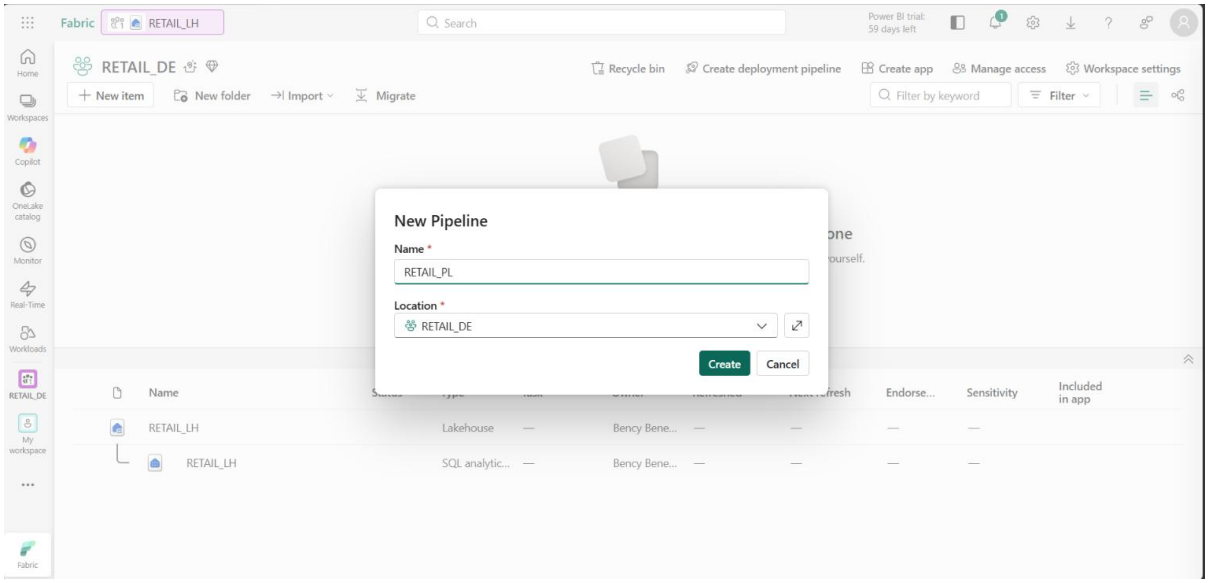
Step 2: Create the Lakehouse

- Open the Fabric workspace **RETAIL_DE** and click New item.
- Create a Lakehouse named **RETAIL_LH** and keep Lakehouse schemas ticked.



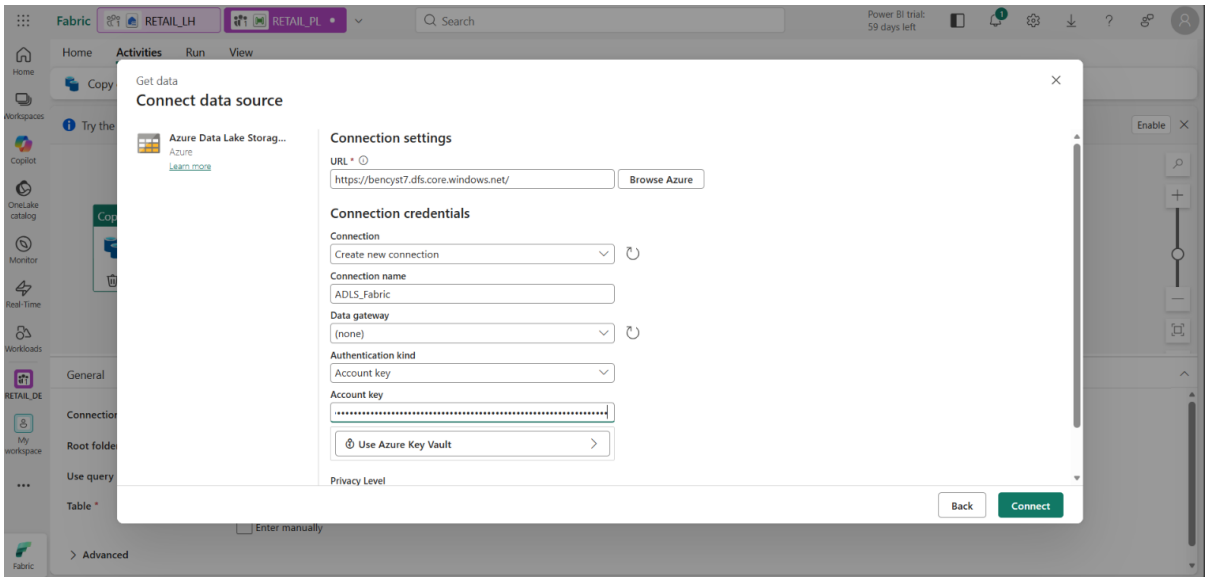
Step 3: Create the Pipeline

- In the same workspace, create a new Data Pipeline named **RETAIL_PL**.



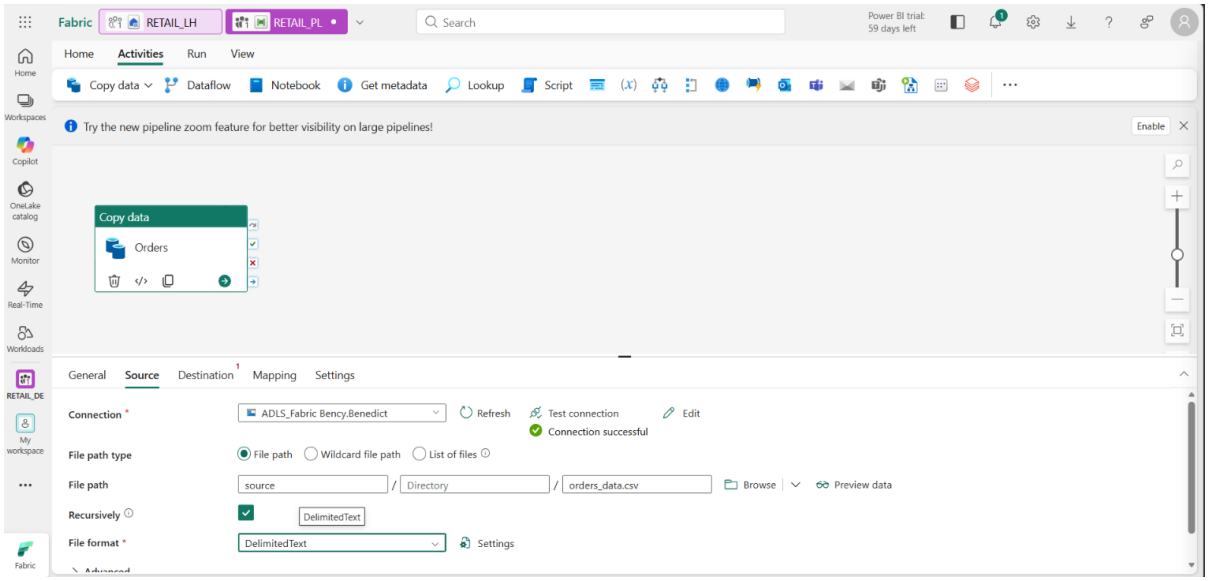
Step 4: Connect to ADLS Gen2

- Add a Copy data activity and choose Azure Data Lake Storage Gen2 as the source.
- Enter the storage account URL, name the connection **ADLS_Fabric**, choose Account key and click Connect.



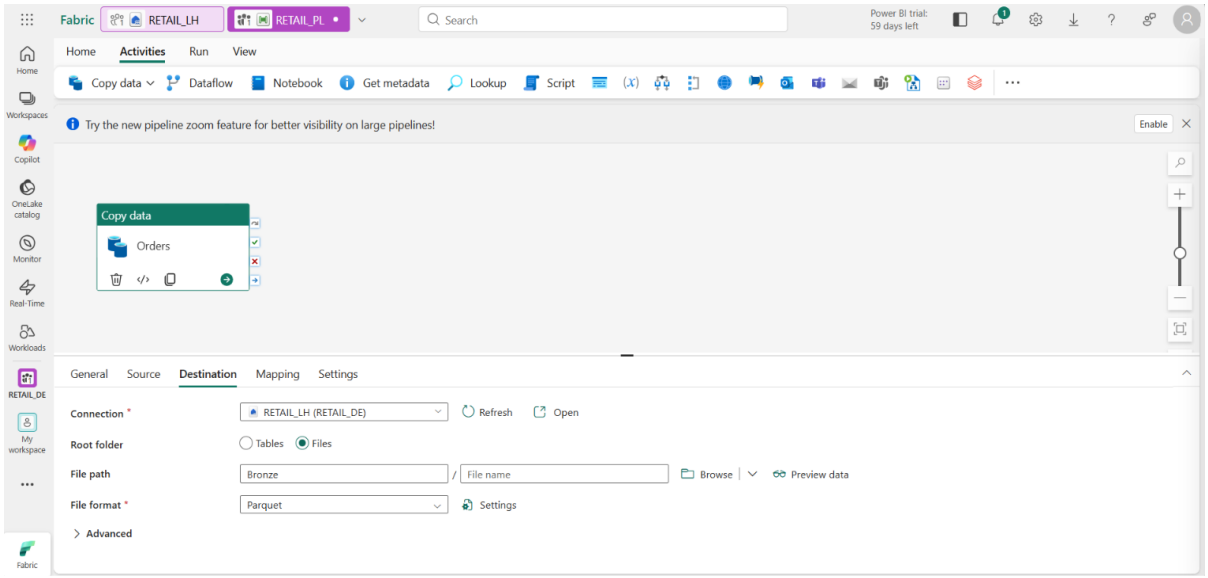
Step 5: Set the Copy Activity Source (Orders)

- Name the activity **Orders**. Select the ADLS_Fabric connection and click Test connection.
- Set the file path to **source / orders_data.csv** and the file format to DelimitedText.



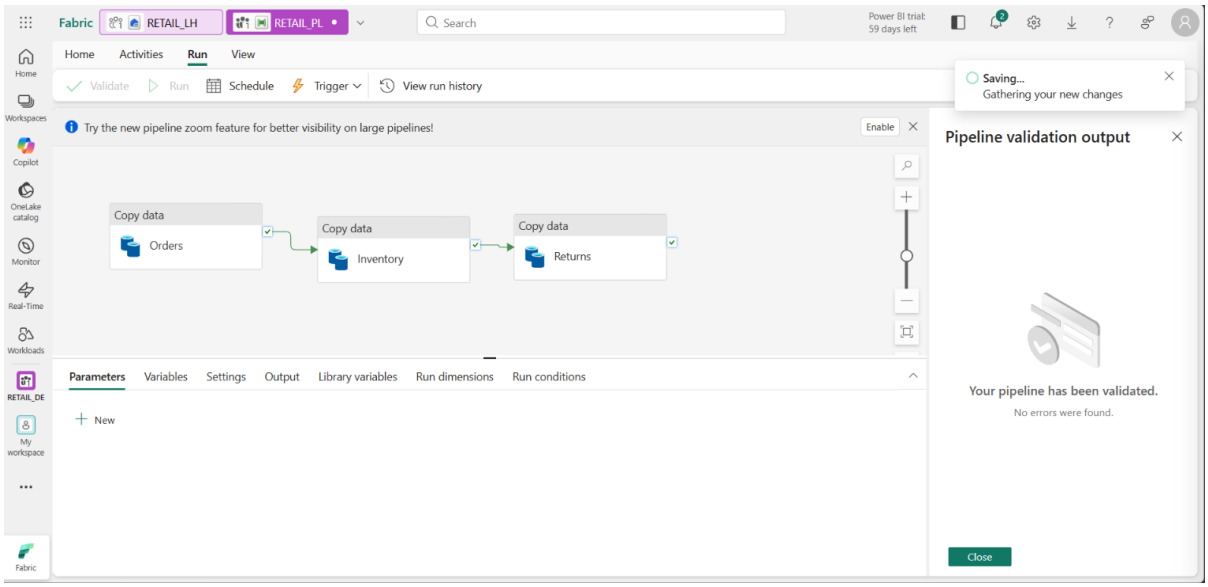
Step 6: Set the Copy Activity Destination (Bronze)

- Choose the **RETAIL_LH** Lakehouse as the destination and select Files as the root folder.
- Set the folder to **Bronze** and the file format to Parquet.



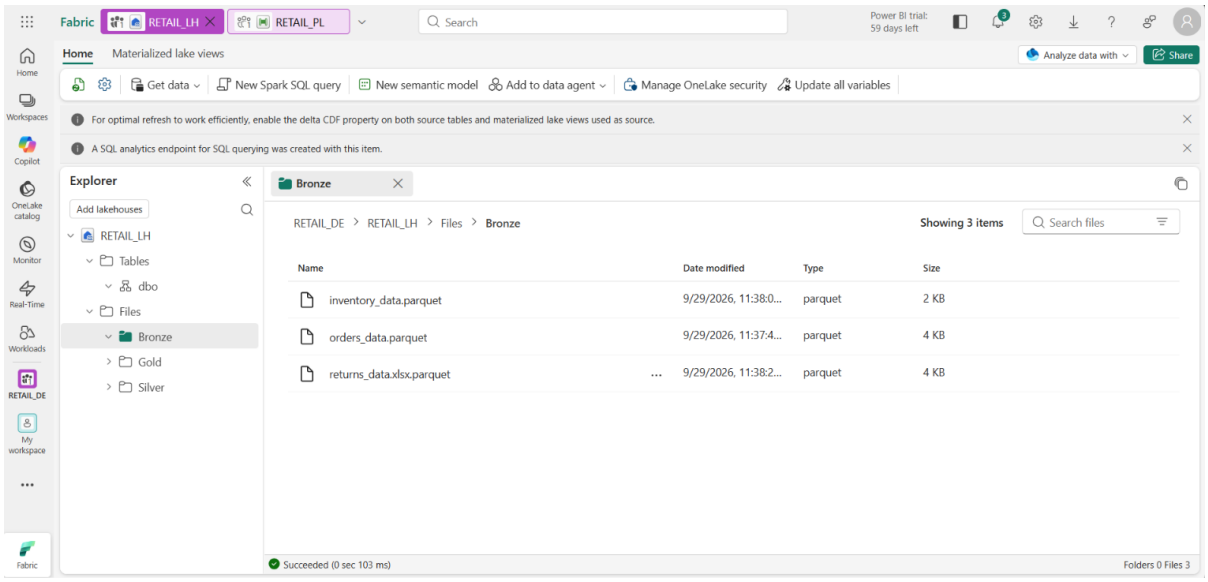
Step 7: Add Inventory and Returns, Then Validate

- Repeat Steps 5 and 6 for **Inventory** (JSON) and **Returns** (Excel).
- Link the three Copy activities in order: Orders, Inventory, Returns. Click Validate. No errors were found.



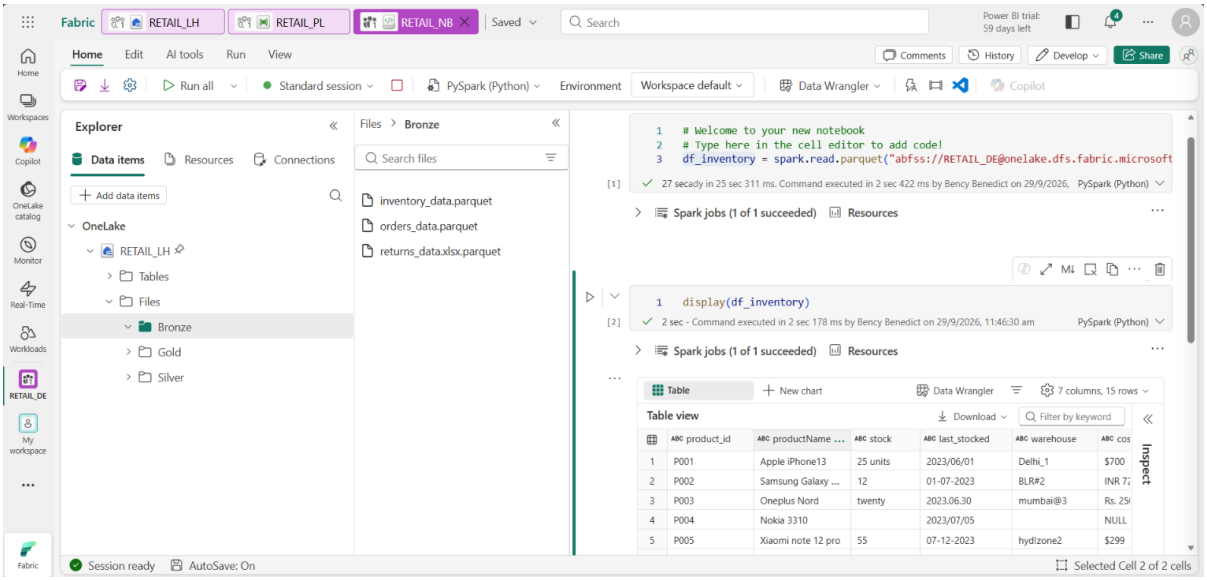
Step 8: Run the Pipeline and Check Bronze

- Run the pipeline and open the Lakehouse.
- The Bronze folder now has three Parquet files: **orders_data**, **inventory_data** and **returns_data**. The Silver and Gold folders are ready for the next layers.



Step 9: Read Bronze Data in a Notebook

- Create a notebook named **RETAIL_NB** and attach the Lakehouse.
- Read the Bronze Parquet file into a Spark DataFrame (**df_inventory**) and display it.



Step 10: Silver Level (Clean and Normalize)

- Read the Bronze data for Orders, Inventory and Returns in the notebook.
- **Orders:** fix column names, convert dates to a proper date format, clean customer IDs, names and emails, fill empty payment modes and promo codes, correct negative amounts, remove rows with missing key values and remove duplicate orders.
- **Inventory:** convert stock to numbers (including text like ‘twenty’), fix dates, keep only the number in cost price, clean warehouse names and change availability to True/False.
- **Returns:** fix column names and dates, clean refund status, amount, address and product text, and remove rows with no return ID.
- Save the cleaned data as Silver tables: **silver_orders**, **silver_inventory** and **silver_returns**.

Step 11: Gold Level (Aggregations and KPIs)

- Read the three Silver tables: orders, returns and inventory.
- Combine them into one table. Each order is matched with its return details and its product stock details. All orders are kept, even if they have no return.
- Group the combined data by product, so we get one row for each product.
- Calculate the KPIs for each product.
- Save the final table.

KPI	What it tells us
Total Orders	How many orders were placed for the product
Unique Customers	How many different customers bought the product
Total Returns	How many orders were returned
Return Rate (%)	Returns divided by orders, shown as a percentage
Total Revenue	Total money earned from the product
Avg Order Value	Average amount of one order
Total Stock	How many units are available in the warehouse
Avg Cost	Average cost price of the product
Net Profit	Revenue minus the cost of the stock